
Algorithm 2 PressureFit planning

Require: $\mathcal{W}$: ordered tasks, object accesses and sizes, runtimes, and initial/final locations

$\mathcal{H}$: memory capacities, directional bandwidths, and deterministic event semantics

Θ : finite implementation-supplied heuristic configurations; each θ fixes

$\text{CutScore}_\theta(g; \mathcal{W}, \mathcal{H}, P, b)$: a totally ordered heuristic key (lower is preferred)

$\text{PrefetchRule}_\theta(\mathcal{W}, \mathcal{H}, P, X) \rightarrow \phi$: one enqueue boundary $\phi(x) \in \text{Window}(x)$ for each prefetch x

Ensure: Fastest simulator-valid annotated plan Γ^* , or $\perp$

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1:  $(\mathcal{B}, \mathcal{A}, q) \leftarrow \text{Analyze}(\mathcal{W})$  ▷ Boundaries, required residency, task-reserved bytes
2:  $\text{FastBytes}_P(b) \equiv q_b + \sum_{o \in \mathcal{O}} s_o \mathbf{1}[b \in P_o]$  ▷ Bytes that must fit at boundary  $b$ 
3:  $(\Gamma^*, m^*) \leftarrow (\perp, \infty)$ 
4: for all  $\theta \in \Theta$  do
5:    $P \leftarrow \{o \mapsto \text{Hull}(\mathcal{A}_o) : o \in \mathcal{O}\}$  ▷ Start with continuous residency
6:   while  $\max_{b \in \mathcal{B}} \text{FastBytes}_P(b) > C_F$  do ▷ Some boundary exceeds capacity
7:     Choose  $b^* \in \mathcal{B}$  with largest  $\text{FastBytes}_P(b)$ 
8:      $G \leftarrow \text{LegalCuts}(P, b^*)$ 
9:     if  $G = \emptyset$  then
10:       $P \leftarrow \perp$ ; break
11:     end if
12:     Choose  $g^* \in G$  with smallest  $\text{CutScore}_\theta(g; \mathcal{W}, \mathcal{H}, P, b^*)$  ▷ Heuristic: which residency gap to remove
13:      $P \leftarrow \text{Cut}(P, g^*)$  ▷ Remove optional residency
14:   end while
15:   if  $P \neq \perp$  then
16:      $X \leftarrow \text{Transitions}(\mathcal{W}, P)$  ▷ Release/offload choices and prefetch windows
17:      $\phi \leftarrow \text{PrefetchRule}_\theta(\mathcal{W}, \mathcal{H}, P, X)$  ▷ Choose one enqueue boundary per prefetch
18:      $\Gamma \leftarrow \text{Emit}(\mathcal{W}, X, \phi)$  ▷ Attach movement annotations to tasks
19:      $(v, m) \leftarrow \text{Simulate}(\Gamma; \mathcal{H})$  ▷ Exact feasibility and makespan
20:     if  $v \wedge m < m^*$  then
21:        $(\Gamma^*, m^*) \leftarrow (\Gamma, m)$ 
22:     end if
23:   end if
24: end for
25: return  $\Gamma^*$ 
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